

Cybersecurity Prevention — Detailed Lecture Notes

JKSSB Computer Awareness

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44.1 Prevention and the CIA triad

Cleaning an infected machine is slow and uncertain, so the first goal of security is prevention: small repeated habits plus a few tools that each do one distinct job. The standard way to judge whether protection is complete is the **CIA triad: confidentiality, integrity, and availability**.

Definition (Confidentiality). *Only authorised persons can read the data. Passwords, encryption, and MFA protect it.*

Definition (Integrity). *The data is accurate and has not been changed by unauthorised means. Clean backups, digital signatures, and timely patching protect it.*

Definition (Availability). *The data and systems are reachable when needed. Backups, antivirus software, and firewalls protect it.*

Term	Meaning in this lesson
Antivirus	Software that detects, blocks, and removes malicious programs.
Signature	A stored pattern of a known virus, used for recognition.
Real-time scanning	Checking files at the moment they are opened, downloaded, or copied.
Quarantine	Holding a suspicious file in an isolated area so it cannot run.
Firewall	A filter that allows or blocks network traffic by rule.
Patch	A small vendor fix for a known security weakness.
MFA — Multi-Factor Authentication	Proving identity with two or more independent factors.
Backup	A separate clean copy of data kept for recovery.
HTTPS	The secure, encrypted form of web browsing.
WPA2/WPA3	Encryption standards that protect a Wi-Fi link.
Autorun	The feature that launches programs automatically when a drive is connected.

One attack can break more than one leg of the triad. Stealing a password over open Wi-Fi breaks confidentiality; altering marks in a result file breaks integrity; ransomware that locks every document breaks availability. The rest of this lesson maps each preventive control to the goal it serves.

44.2 Antivirus: signatures, scanning, and quarantine

Antivirus software detects, blocks, and removes malicious programs (malware). Its oldest method is the **signature**: a stored pattern of a known virus, like a fingerprint kept on record. The

antivirus compares each file against its signature database and flags anything that matches. Because new malware appears every day, the signature database must be updated regularly; an outdated antivirus misses new threats.

Definition (Signature). *A stored pattern identifying a known malicious program, against which files are compared.*

Signatures catch known malware, but they cannot recognise a brand-new (zero-day) threat. Behaviour-based detection watches what a program *does* — for example, trying to encrypt every document — and stops it even when no signature matches.

Two scanning modes matter. **Real-time scanning** checks files at the moment they are opened, downloaded, or copied, before harm is done. A **full (on-demand) scan** checks the whole system when the user asks for it. When either mode finds something suspicious, the antivirus moves the file to **quarantine**: an isolated area where the file cannot run or spread. Quarantine is not deletion. The file is held safely until the user deletes it or restores it (for example, when a clean file was flagged by mistake).

Example 44.2.1. *A pen drive is plugged into the office desktop and a file on it matches a known virus signature. Real-time scanning blocks the file the moment it is opened and moves it to quarantine, so it cannot run. The user then deletes the quarantined file. If instead the file had been a clean office template flagged by mistake, quarantine would allow restoring it — deletion would not.*

44.3 Firewall: the network gatekeeper

A **firewall** monitors incoming and outgoing network traffic and blocks whatever its rules forbid. Its job is to stop unauthorised network access: a stranger on the internet reaching into the office computer, or a malicious program on the computer sending data out. A firewall can be software running on the computer or hardware placed at the network boundary; offices commonly use both.

Definition (Firewall). *A hardware or software filter that allows or blocks network traffic according to a set of rules.*

The firewall–antivirus boundary is a favourite exam trap. A firewall blocks bad *traffic*; it does not remove viruses already saved on the disk. Antivirus removes bad *files*; it does not, by itself, patrol the network boundary. The two complement each other and neither replaces the other.

44.4 Patching and updates

A **patch** is a small fix released by the vendor for a known security weakness, and a **software update** delivers such fixes (plus features). Attackers study published weaknesses, so any delay in installing patches leaves known doors open. The operating system, the web browser, and the antivirus itself must all be updated; each has flaws attackers can exploit. Automatic updates are the safest setting for most users.

Example 44.4.1. *The office desktop offers an operating-system update. Installing it closes a published weakness the same week it became known. Postponing it for months means every attacker who read the same notice knows exactly which door is still open on that machine.*

44.5 Strong passwords and MFA

A **strong password** is long and mixes uppercase letters, lowercase letters, numbers, and symbols. It must be unique per site, never shared, and never left at its factory default: routers and office

devices ship with well-known default passwords that must be changed immediately. A **password manager** stores one distinct strong password per site so the user only memorises a single master secret.

MFA stands for Multi-Factor Authentication. It demands two or more independent proofs of identity: something you know (the password), plus something you have (a one-time code on the phone) or something you are (a fingerprint). Even when the password leaks, the attacker still lacks the second factor, so the account holds. Note the distinction the exam tests: a longer password is still one factor; MFA is the extra check, not a stronger password.

Definition (MFA (Multi-Factor Authentication)). *An authentication method that requires two or more independent factors — for example, a password plus a phone code or fingerprint.*

44.6 Backup: the availability control

A **backup** is a separate clean copy of important data, kept so work can continue when the original is lost, damaged, or locked by ransomware. Backup is the great protector of availability. Good backups are regular (so little work is lost), kept offline or off-site (so one fire, theft, or ransomware attack cannot destroy both copies), and tested by trial restores. A backup that cannot be restored is not a backup.

The backup–antivirus boundary is another tested pair. Backup restores *lost* data; antivirus removes *malicious* data. A machine can have fresh antivirus and still lose a year’s work to a failed disk if there is no backup.

44.7 Safe browsing

Most infections arrive through the browser, so browsing habits are a control in their own right. Look for **HTTPS** and the padlock symbol before entering passwords or bank details; HTTPS encrypts the session between the browser and the site. Treat unexpected links and attachments as hostile: **phishing** messages imitate trusted senders (a bank, an office, a colleague) to steal credentials. Download software only from official or trusted sources, log out on shared computers, and never save passwords on a machine others use.

44.8 Secure Wi-Fi

The wireless link needs its own protection. Home and office networks must use **WPA2 or WPA3** encryption with a strong password, and the router’s default admin password must be changed. Open (password-free) public hotspots expose traffic, so sensitive logins — banking, email, office accounts — should wait for mobile data, a trusted network, or a **VPN** (Virtual Private Network), which encrypts traffic across untrusted links. Two extra habits help: turn off automatic connection to unknown hotspots, and keep software updated so known wireless flaws are patched. Note the layering the exam tests: HTTPS protects the website session, while WPA2/WPA3 protects the wireless link itself; both layers matter.

Definition (VPN (Virtual Private Network)). *An encrypted tunnel that carries traffic safely across an untrusted network.*

44.9 Removable-media safety

USB drives and memory cards carry malware between machines. Scan every drive with antivirus *before* opening its files, and never plug an unknown found drive into the office computer. Disable **autorun** so a newly connected drive cannot launch programs by itself. Handle drives gently at the end as well: use the safe-eject (safely remove hardware) option before pulling the device out, so files are not corrupted.

44.10 Which control does what

This table is the highest-yield page in the lesson; most questions are matching items built from its rows.

Control	Main job
Antivirus + signatures	Detect and remove known malware files.
Real-time scanning + quarantine	Check on access; isolate suspects before harm.
Firewall	Block unauthorised network traffic.
Patching / updates	Close known software weaknesses.
Strong password + MFA	Prove identity; block account theft.
Backup	Restore data after loss or ransomware.
HTTPS + safe browsing	Protect sessions; dodge phishing and bad downloads.
WPA2/WPA3 + VPN	Protect data travelling on the network link.
Scan USB + disable autorun	Stop malware entering on removable drives.

A useful habit: read the stem’s verb. “Remove” points to antivirus, “block traffic” to the firewall, “prove identity” to passwords and MFA, “restore” to backup, and “protect the link” to WPA2/WPA3 or a VPN.

44.11 Exam retrieval and common confusions

Six distinctions prevent most errors on this topic.

1. The firewall blocks **traffic**; the antivirus removes **files**. Do not swap them.
2. Quarantine **isolates**; deletion **removes**. A quarantined file can be restored; a deleted one cannot.
3. MFA adds a **second proof of identity**; a longer password is still one factor.
4. Backup restores **availability**; encryption protects **confidentiality**. A backup is not an antivirus and an antivirus is not a backup.
5. HTTPS protects the **website session**; WPA2/WPA3 protects the **Wi-Fi link**.
6. Confidentiality keeps secrets secret, integrity keeps data correct, and availability keeps systems reachable. Map each attack to the leg it breaks before choosing the control.

Example 44.11.1. *An item asks which control lets the office resume work after ransomware locks every document. Trace the verb: “resume work after loss” means restoring availability, so the answer is the offline backup. Antivirus (removes malware) and the firewall (blocks traffic) cannot unlock or recreate the lost files.*

Exercises

1. State the three goals of the CIA triad in one line each.
2. Explain what a virus signature is and why it must be updated.
3. Distinguish quarantine from deletion.
4. Distinguish a firewall from antivirus software.
5. Explain why a long password alone is not MFA, and what MFA adds.

Solutions

1. Confidentiality: only authorised persons can read the data. Integrity: the data is accurate and unaltered by unauthorised means. Availability: the data and systems are reachable when needed.
2. A signature is a stored pattern of a known virus used for recognition. New malware appears daily, so the database must be updated or new threats will not match anything on record.
3. Quarantine isolates a suspicious file in a safe area where it cannot run, and the file can later be restored; deletion removes the file, and it cannot be recovered from quarantine-style holding.
4. A firewall monitors network traffic and blocks what its rules forbid; it does not clean infected files. Antivirus detects and removes malicious files on the machine; it does not by itself patrol the network boundary.
5. A long password is still a single factor (something you know). MFA (Multi-Factor Authentication) adds at least one independent factor, such as a phone code or fingerprint, so a leaked password alone is not enough.